STATISTICS

Reminding: Normal Distribution

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Rev: 23.10.2022

Normal Distribution

Statistics

In probability theory, a normal distribution is a type of probability density function. Many phenomena in real life (such as the weight of students, and IQ scores of people) can be represented by a normal distribution. A random variable *X* has a normal distribution if its probability density function is given by

$$f(x;\mu,\sigma)=\frac{1}{\sigma\sqrt{2\pi}}e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \quad -\infty<x<\infty \qquad -\infty<\mu<\infty,\ 0<\sigma<\infty$$

where μ is the mean and σ is the standard deviation. To express this situation, we use

$$X\sim N(\mu,\sigma^2)$$

Normal Distribution

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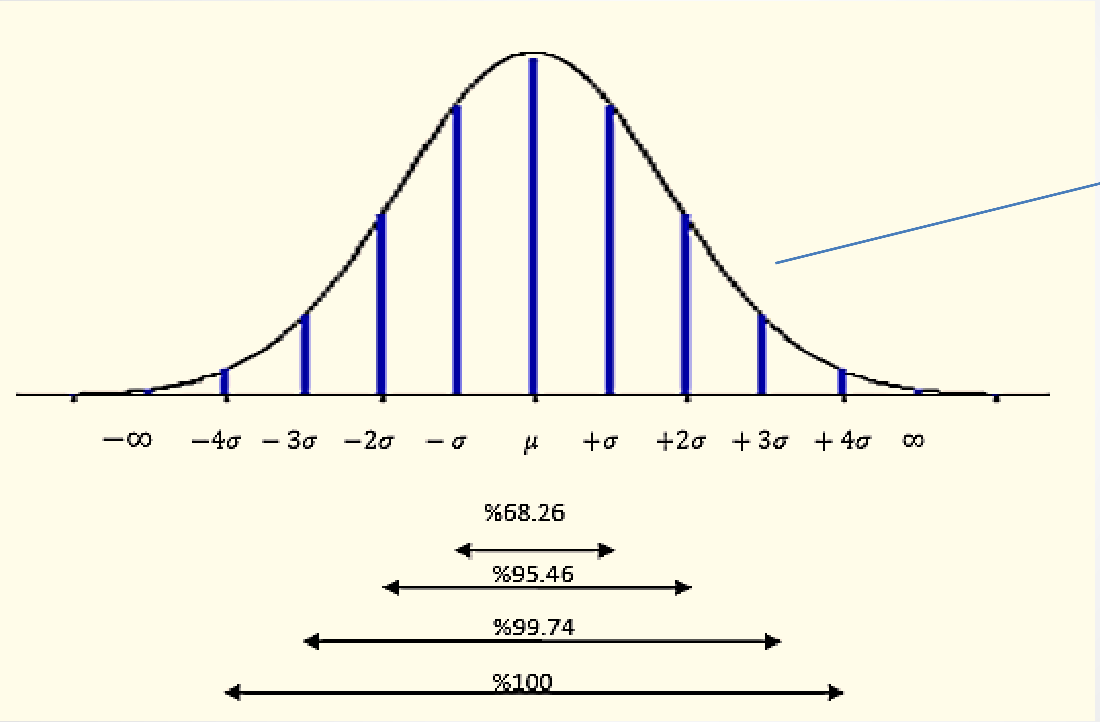

A Curve Bell

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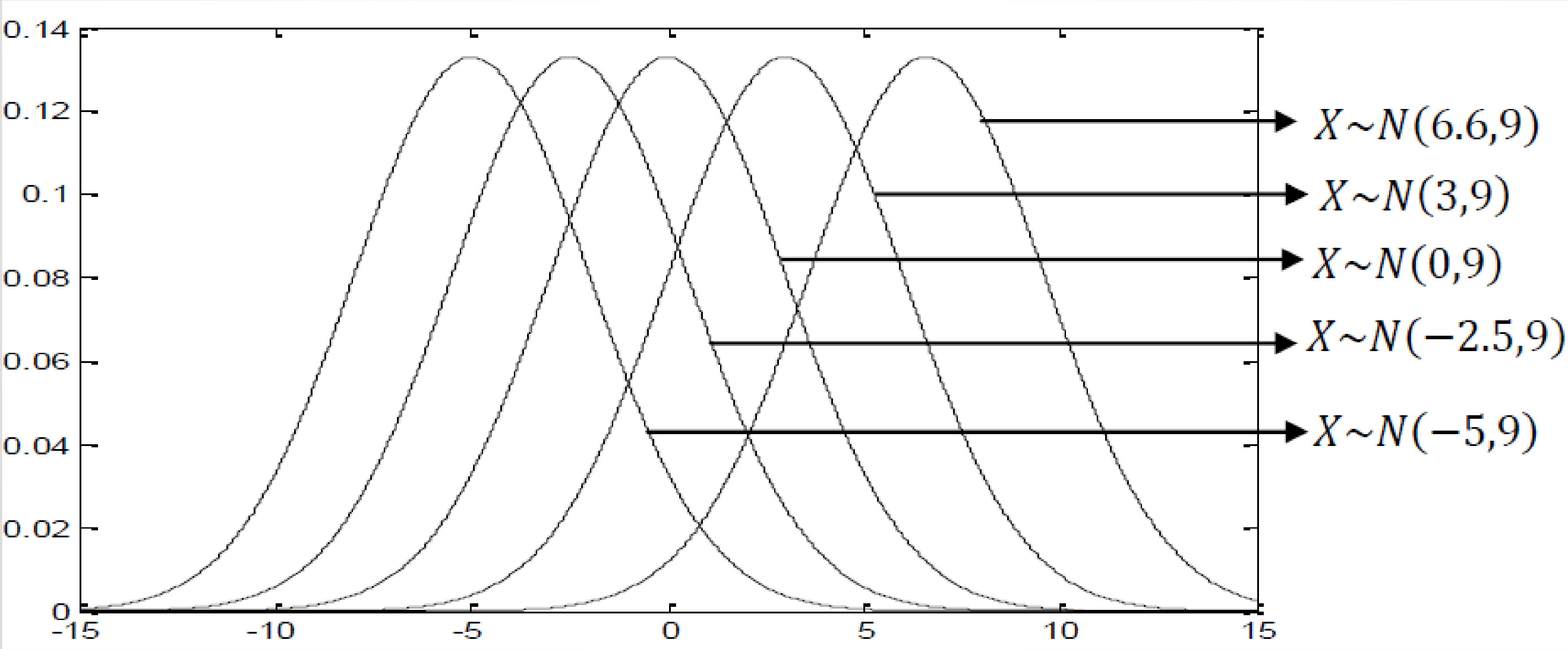

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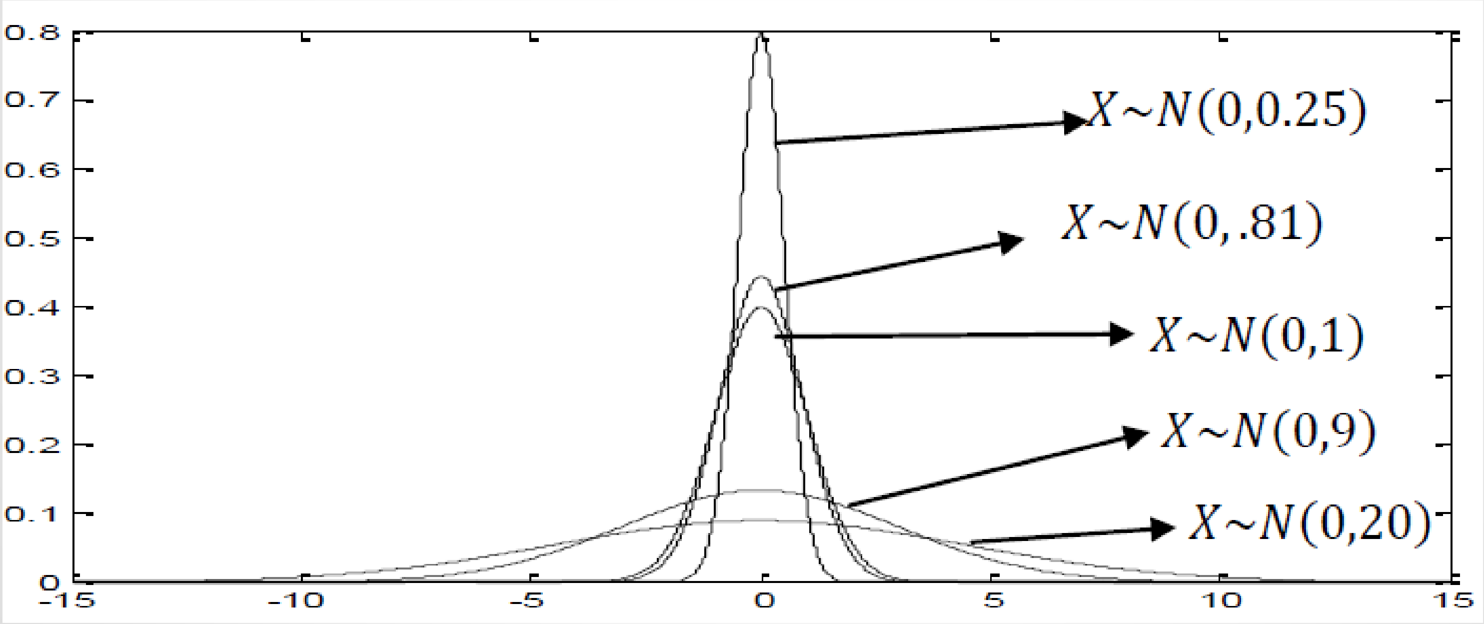

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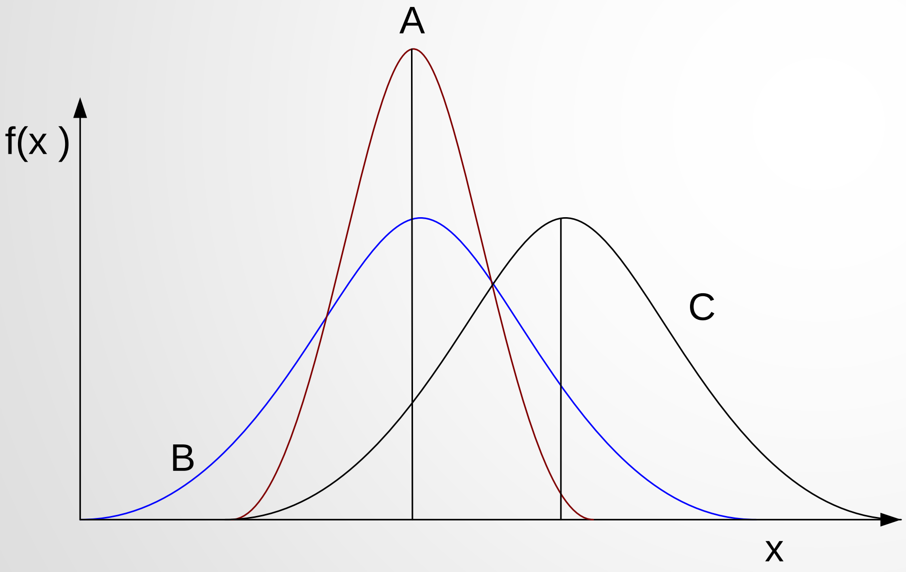

$$\mu_A = \mu_B < \mu_C$$

$$\sigma_A^2 < \sigma_B^2 = \sigma_C^2$$

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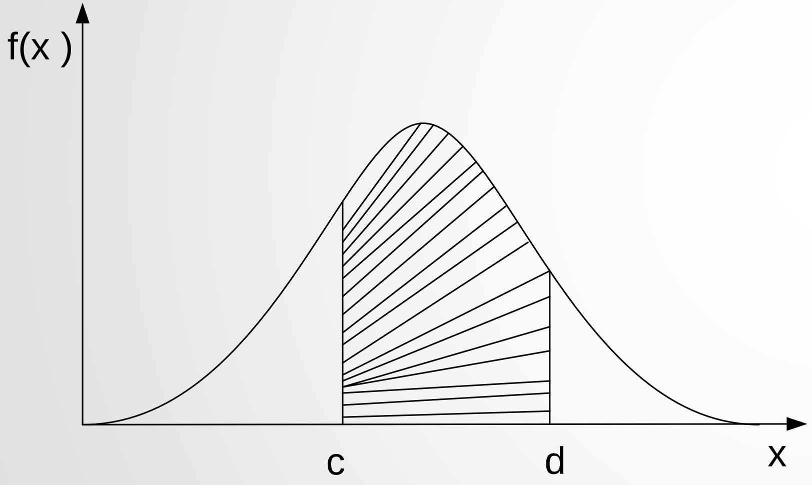

$$P(c \le X \le d) = \int_c^d f(x; \mu, \sigma)\,dx$$

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Standard Normal Distribution

A normal distribution can be transformed into the standard normal distribution by using the standard variable:

$$Z = \frac{X-\mu}{\sigma}$$

In this case;

$$Z = \frac{X-\mu}{\sigma}$$

$$E(Z) = E\left(\frac{X-\mu}{\sigma}\right) = \frac{E(X)-\mu}{\sigma} = \frac{\mu-\mu}{\sigma} = 0,$$

$$Var(Z) = Var\left(\frac{X-\mu}{\sigma}\right) = \frac{Var(X)}{\sigma^2} = \frac{\sigma^2}{\sigma^2} = 1$$

The mean and standard deviation of the standard normal distribution is $\mu = 0$ and $\sigma = 1$, respectively.

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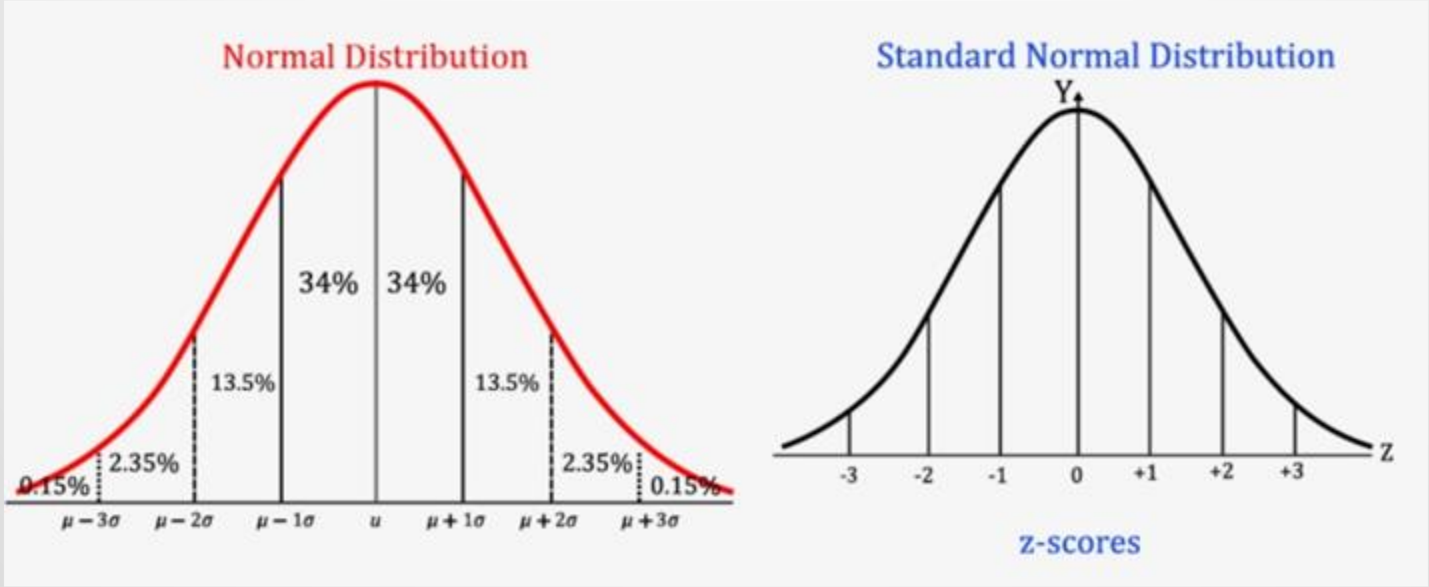
- $X \sim N(\mu, \sigma^2)$
- $Z \sim N(0, 1)$



(1) https://www.pngkit.com/view/u2q8u2e6e6t4y3i1_understanding-the-difference-between-normal-and-standard-normal/

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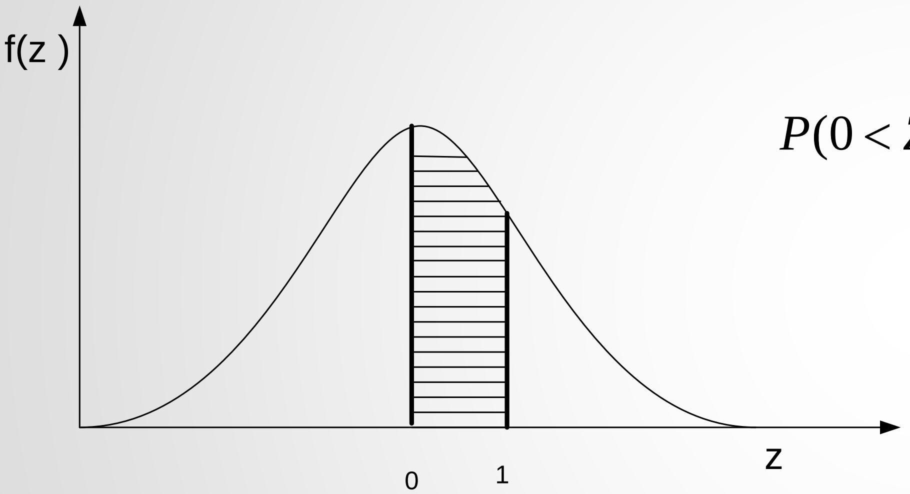
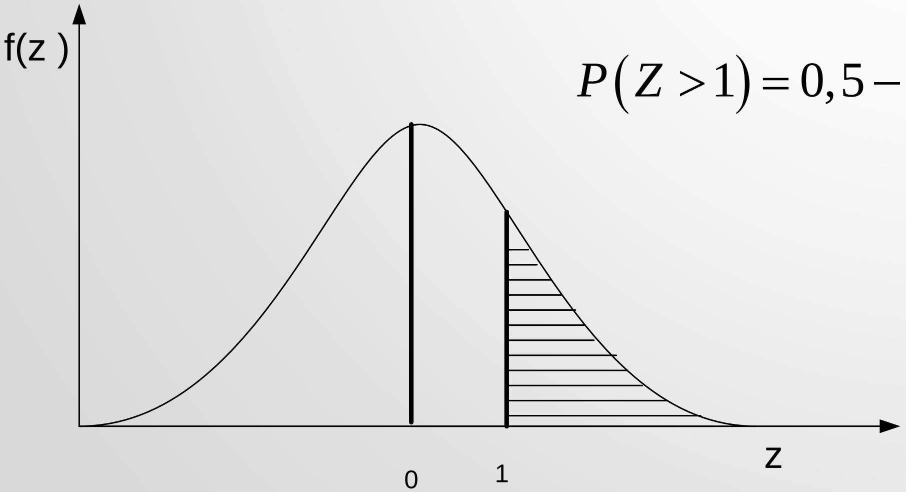


$$P(0 < Z < 1) = 0,3413$$



$$P\left(Z > 1\right) = 0,5 - P(0 < Z < 1) = 0,5 - 0,3413 = 0,1587$$

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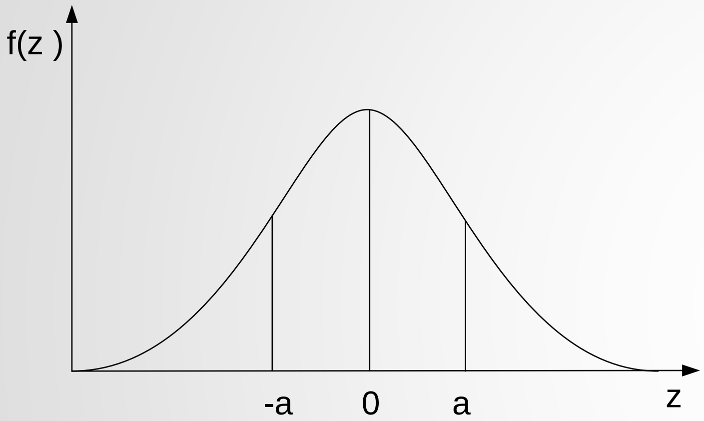
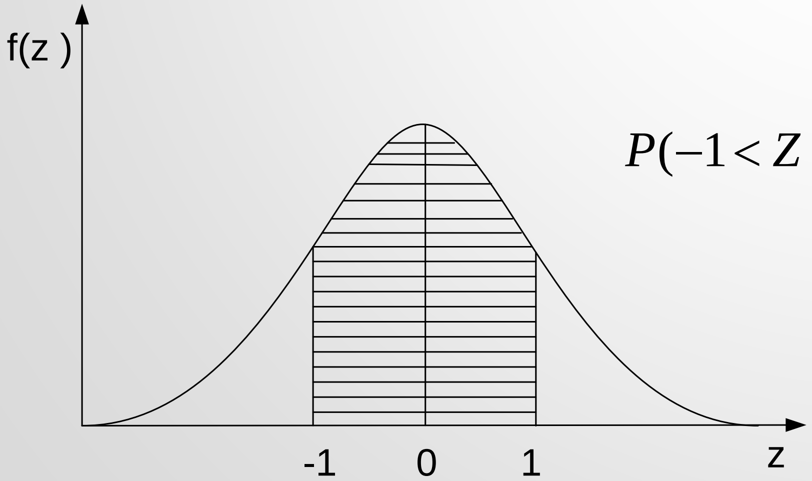


$$P(0 < Z < a) = P(-a < Z < 0)$$



$$P(-1 < Z < 1) = P(-1 < Z < 0) + P(0 < Z < 1)$$
$$= 2P(0 < z < 1) = 2(0,3413) = 0,6826$$

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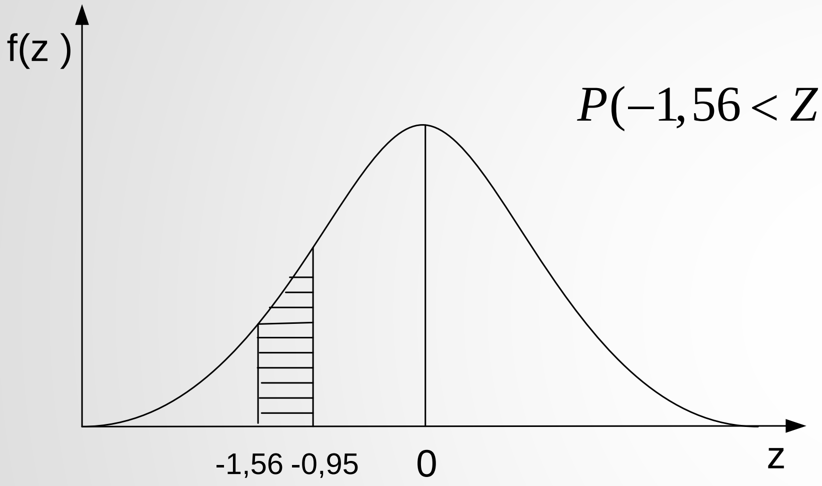
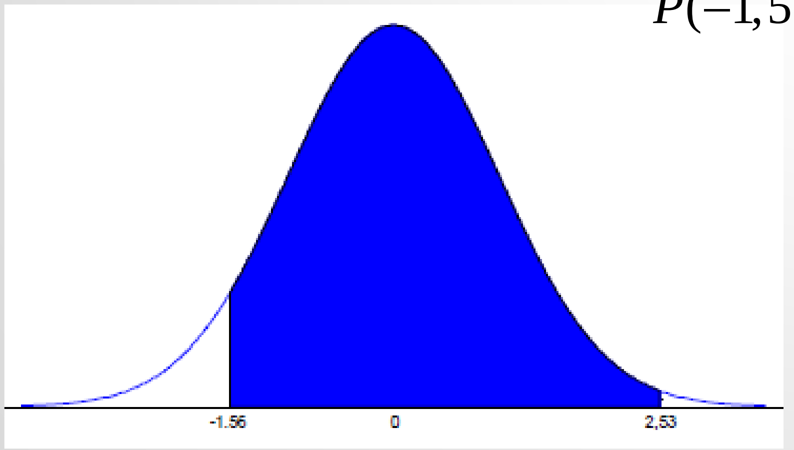


$$P(-1,56 < Z < -0,95) = P(-1,56 < Z < 0) - P(-0,56 < Z < 0)$$
$$= 0,4406 - 0,3289 = 0,1117$$



$$P(-1,56 < Z < 2,53) = P(-1,56 < Z < 0) + P(0 < Z < 2,53)$$
$$= 0,4406 + 0,4942 = 0,9348$$

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Example 1: If X has a normal distribution with the mean $\mu = 3$ and variance $\sigma^2 = 4$, find $P(3 < X < 5)$?

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Example 2: The weight of the students in a class has a normal distribution with the mean $\mu = 68.5$ and standard deviation $\sigma = 2.3$ kg.

a) What is the probability that a randomly chosen student in this class weighs more than 72 kg?
b) What percentage of students' weight is between 70 kg and 72 kg?

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Example 3: The tea boxes with a capacity of 1 kg are filled by machines, and their weight has a normal distribution with the mean $\mu = 1.03$ kg and the standard deviation $\sigma = 0.02$ kg. What is the probability that a random tea box weighs of

a) less than 1 kg?

b) more than 1.06 kg?

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Example 4: Assume that the life of a certain type of battery produced in a factory is normally distributed. If 33% of the batteries last less than 45 hours and 10% of the batteries last more than 65 hours, find the mean and the standard deviation of the distribution.

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Example 5: An exam has a normal distribution with a mean of 75 and a standard deviation of 8. The teacher gives *A* to the students whose grades are 90 or more. If 12 students take *A*, find the number of students attending the exam.